Applied Statistics (MATH10096)

Vanda Inácio

University of Edinburgh

Semester 2, 2024/2025

A simple linear model (Section 2 of the lecture notes)

How old is the universe?

- How old is the universe? Hubble's law, named after Edwin Hubble, is a fundamental principle in physical cosmology that describes the expansion of the universe.
- It states that the recessional velocity of a galaxy (i.e., how fast it is moving away from an observer located on or near Earth) is directly proportional to its distance from Earth.
- In other words, the farther away a galaxy is, the faster it moves away from us. Mathematically:

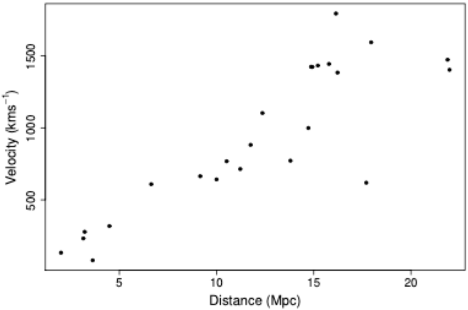
$$y = \beta x.$$

- Here y is the recessional velocity of a galaxy, x is the galaxy's distance from Earth, and β is the 'Hubble constant'.
- Note that β^{-1} gives the approximate age of the universe, but β is unknown and must somehow be estimated from observations of y and x, made for a variety of galaxies at different distances from us.

A simple linear model (Section 2 of the lecture notes)

How old is the universe?

↪ The figure below plots recessional velocity against distance for 24 galaxies, according to measurements made using the Hubble Space Telescope.



A simple linear model (Section 2 of the lecture notes)

How old is the universe?

↪ It is clear, from the figure, that the observed data do not follow Hubble's law exactly, but given the measurement process, it would be surprising if they did.

↪ Given the apparent variability, what can be inferred from these data? In particular:

(i) What value of β is most consistent with the data?

(ii) What range of β values is consistent with the data?

(iii) Are some particular, theoretically derived, values of β consistent with the data?

A simple linear model (Section 2 of the lecture notes)

How old is the universe?

- One way to proceed is to formulate a linear statistical model of the way that the data were generated, and to use this as the basis for inference.

- Specifically, suppose that, rather than being governed directly by Hubble's law, the observed velocity is given by Hubble's constant multiplied by the observed distance plus a 'random variability' term

$$y_i = \beta x_i + \epsilon_i, \quad i = 1 \ldots 24, \tag{1}$$

- The ϵ_i terms are independent random variables such that $\mathbb{E}(\epsilon_i) = 0$ and $\mathbb{E}(\epsilon_i^2) = \sigma^2$, thus implying that $\text{var}(\epsilon_i) = \sigma^2$.

- The random component of the model is intended to capture the fact that if we gathered a replicate set of data, for a new set of galaxies, Hubble's law would not change, but the apparent random variation from it would be different, as a result of different measurement errors.

- Notice that it is not implied that these errors are completely unpredictable: their mean and variance are assumed to be fixed, it is only their particular values, for any particular galaxy, that are not known.

A simple linear model (Section 2 of the lecture notes)

How old is the universe?

- Let's develop statistical methods for a simple linear model of the form (1).
- Formally, consider n pairs of observations, (x_i, y_i), where y_i is an observation on random variable, Y_i, with expectation, $\mu_i \equiv \mathbb{E}(Y_i)$.
- Suppose that an appropriate model for the relationship between x and y is

$$Y_i = \mu_i + \epsilon_i, \quad \text{where} \quad \mu_i = \beta x_i. \tag{2}$$

  Here β is an unknown parameter and the ϵ_i are mutually independent zero mean random variables, each with the same variance σ^2.
- In this regression model we are not including an intercept term, which is the point at which the regression line crosses the y-axis.
- In this case, because we have no intercept the regression line crosses the y-axis at the origin. And this makes sense, as when the distance is zero, the corresponding recessional velocity is also zero.

A simple linear model (Section 2 of the lecture notes)

How old is the universe?

- How can β, in model (2), be estimated from the $\{(x_i, y_i)\}_{i=1}^{n}$ data?
- A sensible approach is to choose a value of β that makes the model fit closely to the data. To do this we need to define a measure of how well, or how badly, a model with a particular β fits the data.
- One possible measure is the residual sum of squares (RSS) of the model

$$\mathcal{S} = \sum_{i=1}^{n} (y_i - \mu_i)^2.$$

- For the current model the RSS is

$$\mathcal{S} = \sum_{i=1}^{n} (y_i - \beta x_i)^2.$$

A simple linear model (Section 2 of the lecture notes)

How old is the universe?

- If we have chosen a good value of β, close to the true value, then the model predicted μ_i should be relatively close to the y_i, so that $\mathcal{S}$ should be small.

- In turn, poor choices will lead to μ_i far from their corresponding y_i, and high values of $\mathcal{S}$.

- Hence β can be estimated by minimizing $\mathcal{S}$ with respect to (w.r.t.) β and this is known as the method of least squares.

- To minimize $\mathcal{S}$, for the current simple model, differentiate w.r.t. β and set the corresponding equation to zero

$$\frac{\partial \mathcal{S}}{\partial \beta} = 0 \Rightarrow \hat{\beta} = \frac{\sum_{i=1}^{n} x_i y_i}{\sum_{i=1}^{n} x_i^2}$$

A simple linear model (Section 2 of the lecture notes)

How old is the universe?

$\hookrightarrow$ Let us consider the sampling properties of $\hat{\beta}$.

$\hookrightarrow$ That is, we should consider some properties of the distribution of $\hat{\beta}$ values, which would be obtained from repeated independent replication of the data used for estimation.

$\hookrightarrow$ To do this, it is helpful to introduce the concept of an estimator, which is obtained by replacing the observations, y_i, in the estimate of $\hat{\beta}$ by the random variables, Y_i, to obtain

$$\hat{\beta} = \sum_{i=1}^{n} x_i Y_i \Big/ \sum_{i=1}^{n} x_i^2.$$

$\hookrightarrow$ Clearly the estimator, $\hat{\beta}$, is a random variable and we can therefore discuss its distribution.

- For now, consider only the first two moments of that distribution.
- The expected value of $\hat{\beta}$ is obtained as follows:

$$\mathbb{E}(\hat{\beta}) = \mathbb{E}\left(\sum_{i=1}^{n} x_i Y_i / \sum_{i=1}^{n} x_i^2\right) = \sum_{i=1}^{n} x_i \mathbb{E}(Y_i) / \sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} x_i^2 \beta / \sum_{i=1}^{n} x_i^2 = \beta.$$

- So $\hat{\beta}$ is an unbiased estimator — its expected value is equal to the true value of the parameter that it is supposed to estimate.

A simple linear model (Section 2 of the lecture notes)

How old is the universe?

- Unbiasedness is a reassuring property, but knowing that an estimator gets it right on average, does not tell us much about how good any one particular estimate is likely to be.
- For this we also need to know how much estimates would vary from one replicate data set to the next — we need to know the estimator variance.
- We have that

$$\mathrm{var}(\hat{\beta}) = \frac{\sum_i x_i^2 \mathrm{var}(Y_i)}{\left(\sum_i x_i^2\right)^2} = \frac{\sigma^2}{\sum_i x_i^2}. \quad (3)$$

- Note that $\mathrm{var}(\hat{\beta})$ depends on σ^2 which is also, in most circumstances, an unknown parameter and therefore it must also be estimated.

- Since σ^2 is the variance of the ϵ_i, it makes sense to estimate it using the variance of the 'estimated' ϵ_i, the model residuals.

- In general these are defined as $\hat{\epsilon}_i = y_i - \hat{\mu}_i$, where $\hat{\mu}_i$ are the fitted values, the model estimates of $\mu_i = \mathbb{E}(y_i)$.

- For the current simple model $\hat{\mu}_i = \hat{\beta} x_i$, so $\hat{\epsilon}_i = y_i - \hat{\beta} x_i$.

- An unbiased estimator of σ^2 for this model is then:

$$\hat{\sigma^2} = \frac{1}{n-1} \sum_i (y_i - \hat{\beta} x_i)^2 \tag{4}$$

- Plugging this estimate of σ^2 into (3) obviously gives an unbiased estimate of the variance of $\hat{\beta}$.

A simple linear model (Section 2 of the lecture notes)

How old is the universe?

↪ Let us implement our simple model in `R` through the `lm` function.

```
library(gamair); data(hubble)
#To know more about the data
help(hubble)

#Fit simple linear regression model (with no intercept)
hub.mod <- lm(y ~ x - 1, data = hubble)
summary(hub.mod)$coefficients
```

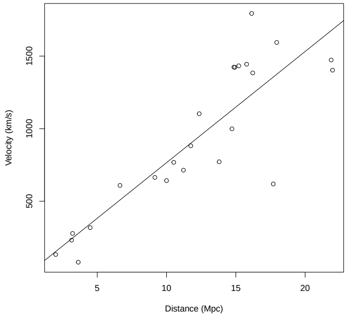
```
##   Estimate Std. Error t value     Pr(>|t|)
## x 76.58117   3.964794 19.3153 1.031907e-15
```

A simple linear model (Section 2 of the lecture notes)

How old is the universe?

↪ Recall that the 'fitted values' for this model are defined as $\hat{\mu}_i = \hat{\beta} x_i$.

```
plot(hubble[,3], hubble[, 2],
     xlab = "Distance (Mpc)", ylab = "Velocity (km/s)")
abline(hub.mod)
```

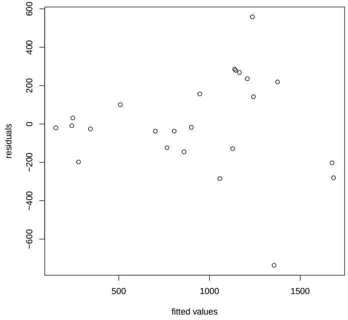


A simple linear model (Section 2 of the lecture notes)

How old is the universe?

- For this model the residuals are simply $\hat{\epsilon}_i = y_i - \hat{\mu}_i = y_i - \hat{\beta} x_i$.
- A plot of residuals against fitted values is often useful to check the model assumptions.

```
plot(fitted(hub.mod), residuals(hub.mod),
     xlab="fitted values", ylab="residuals")
```



- What we would like to see, in such a plot, is an apparently random scatter of residuals around zero, with no trend in either the mean of the residuals, or their variability, as the fitted values increase.
- The main problematic feature of the figure in the previous slide is the presence of two points with very large magnitude residuals, suggesting a problem with the constant variance assumption.
- It is probably prudent to repeat the model fit, with and without these points, to check that they are not having undue influence on our conclusions.

A simple linear model (Section 2 of the lecture notes)

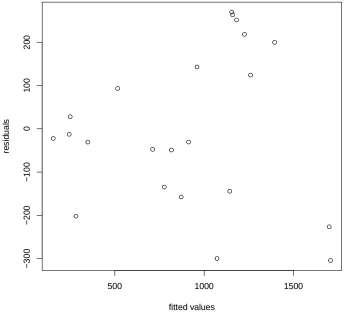
How old is the universe?

```
#Extracting the index of the outlying observations
as.numeric(which(residuals(hub.mod) > 400 | residuals(hub.mod) < -600))

## [1]  3 15

#Fitting model without outliers
hub.mod1 <- lm(y ~ x - 1, data = hubble[-c(3, 15), ])
#Another way of extracting the coefficients from the summary output
coef(summary(hub.mod1))

##   Estimate Std. Error  t value     Pr(>|t|)
## x 77.67292    2.97028 26.15003 1.659996e-17
```

A simple linear model (Section 2 of the lecture notes)

How old is the universe?

```r
plot(fitted(hub.mod1), residuals(hub.mod1),
     xlab="fitted values", ylab="residuals")
```



↪ The omission of the two large outliers has improved the residuals and changed $\hat{\beta}$ somewhat, but not drastically

A simple linear model (Section 2 of the lecture notes)

How old is the universe?

- The Hubble constant estimates have units of $(\mathrm{km})\mathrm{s}^{-1}\ (\mathrm{Mpc})^{-1}$.
- A Mega-parsec is $3.09 \times 10^{19}\mathrm{km}$, so we need to divide $\hat{\beta}$ by this amount, in order to obtain Hubble's constant with units of s^{-1}.
- The approximate age of the universe, in seconds, is then given by the reciprocal of $\hat{\beta}$.
- Here are the two possible estimates expressed in years:

```
hubble.const <- c(coef(hub.mod), coef(hub.mod1))/3.09e19
#Approximate age of the universe in seconds
age <- 1/hubble.const
#Approximate age of the universe in years
age/(60^2*24*365)

##           x           x
## 12794692825 12614854757
```

- Both fits give an age of around 13 billion years.

A simple linear model (Section 2 of the lecture notes)

How old is the universe?

- So far everything done with the simple model has been based only on the model equations and the two assumptions of independence and equal variance of the error terms.
- If we wish to go further, and find confidence intervals for β, or test hypotheses related to the model, then a further distributional assumption will be necessary.
- Specifically, assume that $\epsilon_i \sim N(0, \sigma^2)$ for all i, which is equivalent to assuming $Y_i \sim N(\beta x_i, \sigma^2)$.
- Now we have already seen that $\hat{\beta}$ is just a weighted sum of Y_i, but the Y_i are now assumed to be normal random variables, and a weighted sum of normal random variables is itself a normal random variable.
- Hence,

$$\hat{\beta} \sim N\left(\beta, \left(\sum x_i^2\right)^{-1} \sigma^2\right). \tag{5}$$

- One thing we might want to do is to try and evaluate the consistency of some hypothesized value of β with the data.

- For example some Creation Scientists estimate the age of the universe to be 6000 years, based on a reading of the Bible, thus implying that $\beta = 163 \times 10^6$.

- The consistency with data of such a hypothesized value for β can be based on the probability that we would have observed the $\hat{\beta}$ actually obtained, if the true value of β was the hypothetical one.

- Specifically, we can test the null hypothesis, $\mathrm{H}_0 : \beta = \beta_0$, versus the alternative hypothesis, $\mathrm{H}_1 : \beta \neq \beta_0$, for some specified value β_0, by examining the probability of getting the observed $\hat{\beta}$, or one further from β_0, assuming H_0 to be true.

- The probability required is known as the p-value of the test. It is

  the probability of obtaining a value of $\hat{\beta}$ at least as extreme (i.e., at least as favourable to H_1) as the one actually observed, if H_0 is actually true.

- In this case it helps to distinguish notationally between the estimate, $\hat{\beta}_{\text{obs}}$, and estimator $\hat{\beta}$. The p-value is then

$$
\begin{aligned}
p &= \Pr\left[|\hat{\beta} - \beta_0| \geq |\hat{\beta}_{\text{obs}} - \beta_0| \,\middle|\, \mathrm{H}_0 \right] \\
&= \Pr\left[\frac{|\hat{\beta} - \beta_0|}{\sigma_{\hat{\beta}}} \geq \frac{|\hat{\beta}_{\text{obs}} - \beta_0|}{\sigma_{\hat{\beta}}} \,\middle|\, \mathrm{H}_0 \right] \\
&= \Pr[|Z| > |z|]
\end{aligned}
$$

- Here $Z \sim N(0,1)$, $z = (\hat{\beta}_{\text{obs}} - \beta_0)/\sigma_{\hat{\beta}}$ and $\sigma^2_{\hat{\beta}} = (\sum x_i^2)^{-1}\sigma^2$.

- Small p-values suggest that the data are inconsistent with H_0, while large values do not provide evidence against H_0.

- The threshold 0.05 is often used as the boundary between ‘small’ and ‘large’ in this context.

A simple linear model (Section 2 of the lecture notes)

How old is the universe?

- In reality σ^2 is usually unknown.
- Broadly the same testing procedure can still be adopted, by replacing σ with $\hat{\sigma}$ (see Equation (4)), but we need to somehow allow for the extra uncertainty that this introduces (unless the sample size is very large).
- It turns out that if $\mathrm{H}_0 : \beta = \beta_0$ is true then

$$T \equiv \frac{\hat{\beta} - \beta_0}{\hat{\sigma}_{\hat{\beta}}} \sim t_{n-1}.$$

- We can calculate a p-value by evaluating

$$p = \mathrm{Pr}[|T| > |t|].$$

where $T \sim t_{n-1}$ and $t = (\hat{\beta}_{\mathrm{obs}} - \beta_0)/\hat{\sigma}_{\hat{\beta}}$.

A simple linear model (Section 2 of the lecture notes)

How old is the universe?

↪ Here is some code to evaluate the p-value for H_0 : *the Hubble constant is 163000000*.

```
cs.hubble <- 163000000
#test statistic
t.stat <- (coef(hub.mod1) - cs.hubble)/summary(hub.mod1)$coefficients[2]
t.stat

##         x
## -54876951

#p-value
df <- nrow(hubble[-c(3, 15), ]) - 1
df

## [1] 21

pt(abs(t.stat), df = df, lower = FALSE)*2

##             x
## 3.906388e-150
```

A simple linear model (Section 2 of the lecture notes)

How old is the universe?

- Having seen how to test whether a *particular* hypothesized value of β is consistent with the data, the question naturally arises of what range of values of β would be consistent with the data?
- To answer this, we need to select a definition of 'consistent': a common choice is to say that any parameter value is consistent with the data if it results in a p-value of ≥ 0.05, when used as the null value in a hypothesis test.
- Sticking with the Hubble constant example, and working at a significance level of 0.05, we would have rejected any hypothesized value for the constant, that resulted in a *t* value outside the range $(-2.08, 2.08)$, since these values would result in p-values of less than 0.05.
- So we would not reject any β_0 fulfilling:

$$-2.08 \leq \frac{\hat{\beta} - \beta_0}{\hat{\sigma}_{\hat{\beta}}} \leq 2.08$$

which re-arranges to give the interval

$$\hat{\beta} - 2.08\hat{\sigma}_{\hat{\beta}} \leq \beta_0 \leq \hat{\beta} + 2.08\hat{\sigma}_{\hat{\beta}}.$$

Such an interval is known as a '95% Confidence interval' for β.

A simple linear model (Section 2 of the lecture notes)

How old is the universe?

- The defining property of a 95% confidence interval is this: if we were to gather an infinite sequence of independent replicate data sets, and calculate 95% confidence intervals for β from each, then 95% of these intervals would include the true β, and 5% would not.

- It is easy to see how this comes about. By construction, a hypothesis test with a significance level of 5% rejects the correct null hypothesis for 5% of replicate data sets, and does not reject it for the other 95% of replicates. Hence 5% of 95% confidence intervals must exclude the true parameter, while 95% include it.

A simple linear model (Section 2 of the lecture notes)

How old is the universe?

- For the Hubble example, a 95% CI for the constant (in the usual astro-physicists units) is given by

```
sigb <- summary(hub.mod1)$coefficients[2]
h.ci <- coef(hub.mod1) + qt(c(0.025, 0.975), df = df)*sigb
h.ci
```

```
## [1] 71.49588 83.84995
```

- This can be converted to a confidence interval for the age of the universe, in years, as follows

```
h.ci <- h.ci*60^2*24*365.25/3.09e19 # convert to 1/years
sort(1/h.ci)
```

```
## [1] 11677548698 13695361072
```

- That is, the 95% CI is (11.7,13.7) billion years. Actually this 'Hubble age' is the age of the universe if it has been expanding freely, neither being slowed down by gravitation, nor accelerating for reasons not fully understood.

- In fact this free expansion assumption does not appear to be quite right, so our answer will be out by a bit more than the CI might suggest.